

Richard Motorgeanu

motorger@mcmaster.ca | multipixels.dev | github.com/Multipixels

Education

McMaster University – Hamilton, ON

Expected May 2027

Bachelor of Applied Science in Computer Science

GPA: 4.00/4.00

Minor in Mathematics

Courses: Advanced Calculus, Computer Graphics, Cryptography, Data Privacy, Natural Language Processing, Physics for Video Games, Probability and Games of Chance

Experience

Learning Resource Technical Assistant

May 2024 – Present

McMaster University – Hamilton, ON

- Designed and developed a README generator tool in JavaScript to streamline the creation of README files, helping researchers document their datasets
- Automated the process of querying the Echo360 API using Python and GitHub Actions to filter out video content created by the library and aggregate relevant statistics for over 300 workshop videos
- Created a search tool with JavaScript that allows users to browse and discover over 100 workshops

Learning Resource Development Assistant

May 2023 – Apr 2024

McMaster University – Hamilton, ON

- Redesigned workshop websites hosted on GitHub Pages to enhance user experience, ensuring compliance with web accessibility standards and improving overall navigation
- Collaborated with team members to develop learning material from previous workshops into asynchronous learning modules, tailoring content to meet the needs of learners within and beyond McMaster University

Teaching Assistant

June 2022 – July 2022

Peel District School Board – Mississauga, ON

- Mentored students through the development of their final project by offering constructive feedback and troubleshooting technical challenges, resulting in 30+ unique projects

Projects

Lotus Programming Language

github.com/Multipixels/lotus-lang

- Designed and developed Lotus, an interpreted programming language, with a custom interpreter built in C++ to enhance my understanding of language design
- Implemented static typing and error checking to improve language functionality and developer experience
- Employed test-driven development with GoogleTest to ensure the reliability and correctness of Lotus's interpreter and core features

Nyctomorph

multipixels.dev/projects/nyctomorph/

- Worked collaboratively to develop a 2D survival adventure game in Godot in just one week for a game jam
- Achieved top 10 rankings for ranked for quality and top 20 overall among more than 125 entries

Keep Talking and Nobody Explodes Bot

multipixels.dev/projects/ktane-assistant/

- Created a voice assistant using Python to assist players in solving puzzles in Keep Talking and Nobody Explodes
- Documented the bot's code and created an installation and usage guide to promote sustainable development

Skills

Languages: Python, C#, C++, C, Java, HTML/CSS/JS, Sass, Go, SQL, Haskell, R

Tools and Frameworks: OpenGL, Unity, Godot, Git, GitHub, GitHub Pages, Jekyll, VSCode, Visual Studio, Microsoft 365, AutoCAD, GoogleTest